

ABNORMALITY REPORT — RCA & CAPA/PAA

Equipment Related Risk

Product Blower BL-5702

High Vibration (Coupling Misalignment)

AR NUMBER	AR-2026-OPP-0203
TAG / PLANT	BL-5702 OPP
DATE OCCURRENCE	17 Jun 2026
PRE-RISK	III (Score 400)
PIC (RCA)	ROT-01

Procedure No. RCA-P-0050-03 • Form RCA-F-0075-02

AR Details & Problem Statement

AR Identification

Title / Problem	Product Blower BL-5702 — High Vibration (Coupling Misalignment)
AR Type	OPEDR (Operation) — Uptime
Plant / Unit	OPP (Orion Polypropylene Plant)
Tag / Eq. Class	BL-5702 Class A
Discipline	ROT
Date Occurrence	17 Jun 2026
Date Reported	18 Jun 2026
Immediate Action	Stop BL-5702, re-align coupling, replace worn coupling element

Severity & Impact

SEVERITY	PRE-RISK	DOWNTIME	LOSS
Major	III	14.0 h	\$478.8k

Problem Statement

Product Blower BL-5702 tripped on high vibration on 17-Jun-2026 due to coupling misalignment and a worn coupling element, halting the OPP powder-handling section.

Chronology of Events

Date / Time	Event
15-Jun-2026	OPP running normal; BL-5702 vibration trending up (4.2 → 8.5 mm/s) with 2X harmonic rising.
17-Jun-2026 04:30	BL-5702 vibration reached alarm (11 mm/s); blower stopped to prevent damage — powder handling section halted.
17-Jun-2026 07:00	Coupling element found worn/cracked; laser alignment showed 0.35 mm offset (spec < 0.05 mm).
17-Jun-2026 18:30	Coupling element replaced, unit re-aligned to < 0.05 mm, blower restarted to 38 T/H.

Problem: Product Blower BL-5702 tripped on high vibration on 17-Jun-2026 due to coupling misalignment and a worn coupling element, halting the OPP powder-handling section.

Past Performance Analysis

Historical Data & Evidence

BL-5702 vibration baseline is 3.5–4.5 mm/s. A rising trend with dominant 2X harmonic over 2 days indicated developing misalignment. Alignment was last verified 14 months prior; no soft-foot / alignment check in the routine PM.

Impact Summary (4W)

What	BL-5702 high vibration from coupling misalignment
When	17-Jun-2026 (04:30)
Scope	OPP powder-handling section stop
Uptime Loss	14.0 hours → 532.0 ton → \$478.8k

Actual vs Target Condition

Actual Condition (Before Improvement)

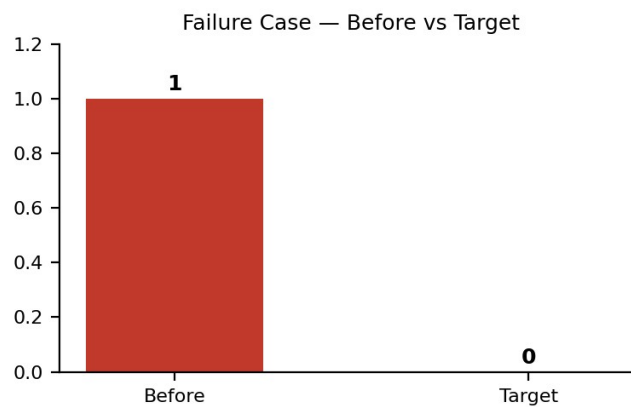
Blower stop due to high vibration (misalignment)

Target Condition (Project Y)

No blower stop due to misalignment (vibration < 7 mm/s, alignment < 0.05 mm)

Target Metric

Recurrence of this equipment failure mode: Before = 1 case → Target = 0 case



Parameter Verification (4P)

#	Problem Phenomenon / Parameter	Result	Evidence / Finding
P1	Overall vibration above alarm	NG	Vibration reached 11 mm/s vs 7 mm/s alarm — misalignment signature.
P2	Coupling alignment out of tolerance	NG	Offset 0.35 mm vs < 0.05 mm spec — angular/parallel misalignment.
P3	Bearing condition degraded	G	Bearing envelope/shock-pulse normal — no bearing defect.
P4	Foundation / soft-foot	NG	Soft-foot 0.12 mm found on one foot — contributed to misalignment.
P5	Rotor unbalance	G	1X component within balance grade — no unbalance.

NG = current condition not meeting standard (problem) • G = meets standard

4M + 1E Verification

#	Man / Machine / Method / Material / Environment	Result	Evidence / Finding
X1	Alignment check not in routine PM	NG	PM did not include periodic laser alignment / soft-foot check.
X2	Coupling element life not tracked	NG	Elastomer coupling element run beyond recommended 12-month life.
X3	Vibration route interval too long	NG	Monthly route missed the fast-rising 2-day trend.
X4	Alignment tools / competency	G	Laser alignment tool & trained technicians available.

ROOT CAUSE: High vibration from coupling misalignment aggravated by soft-foot and an over-aged elastomer coupling element, undetected because periodic alignment/soft-foot checks were absent from the PM and the vibration route interval was too long.

Impact vs Control Prioritization

Matrix: Control (rows) vs Impact (columns)

CONTROL ↓ / IMPACT →	Low	Medium	High
Low			
Medium			X1 X2 X3
High			P1 P2 P4

Prioritized root causes (high impact) are addressed first via Corrective & Pro-Active Actions →

Corrective & Pro-Active Action (CAPAA)

Corrective Action

RC	Corrective Action	Plan Date	PIC	Status
X1	Add periodic laser alignment & soft-foot check to BL-5702 PM	18-Jul-2026	ROT-01	In Progress
X2	Track coupling element life & replace at 12 months	18-Jul-2026	REL-05	Open

Pro-Active Action (roll-out to similar equipment)

RC	Pro-Active Action	Plan Date	PIC	Status
X1	Roll out alignment check to all Class-A blowers in OPP	18-Aug-2026	ROT-01	Open
X2	Create coupling-element replacement register for OPP trains	31-Aug-2026	REL-05	Open

Preventive & Risk Analysis

Preventive / Pro-Active Action (Good Condition Parameters)

Item	Possible Root Cause	Preventive Action	Plan Date	PIC
X3	Route interval too long	Shorten BL-5702 vibration route to weekly	18-Jul-2026	REL-05
P4	Soft-foot	Correct soft-foot & re-shim baseplate BL-5702	18-Jun-2026	ROT-01

Risk Analysis of Corrective Action

Corrective Action	Potential Risk	Countermeasure	Plan Date	PIC
Shorten vibration route to weekly	Increased inspection workload	Use wireless vibration sensor for continuous trend	18-Jul-2026	REL-05

PM Schedule Established

PM No.	Description	Group	Interval
PM-1	6-monthly laser alignment & soft-foot check BL-5702	Rotating	6-monthly
PM-2	Coupling element inspection/replacement	Rotating	12-monthly
PM-3	Weekly vibration trend review BL-5702	Reliability	Weekly

Downtime & Closure Summary

DOWNTIME	PRODUCTION LOSS	ESTIMATED LOSS	PRE-RISK
14.0 h	532.0 ton	\$478.8k	III (400)

Root Cause

High vibration from coupling misalignment aggravated by soft-foot and an over-aged elastomer coupling element, undetected because periodic alignment/soft-foot checks were absent from the PM and the vibration route interval was too long.

Key Corrective / Pro-Active Actions

- Add periodic laser alignment & soft-foot check to BL-5702 PM
- Track coupling element life & replace at 12 months
- Shorten BL-5702 vibration route to weekly